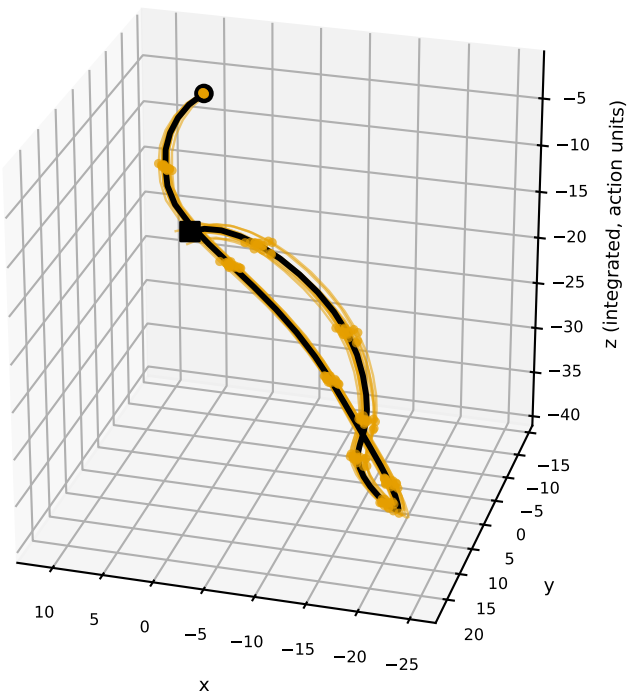
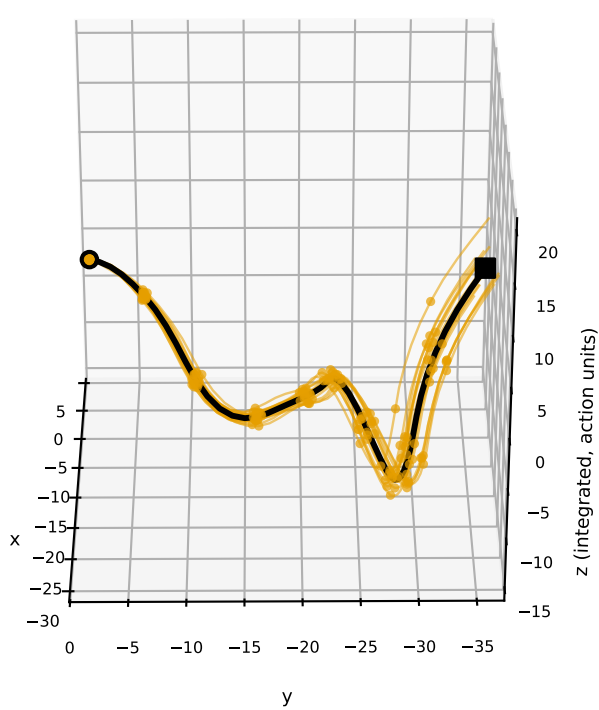


$\pi_{0.5}$ flow on LIBERO. Top: sampled chunks as 3-D paths at three states. Bottom: the 'two clouds' at the most separated state are the gripper's binary flip landing one step early or late; the arm does not know which.

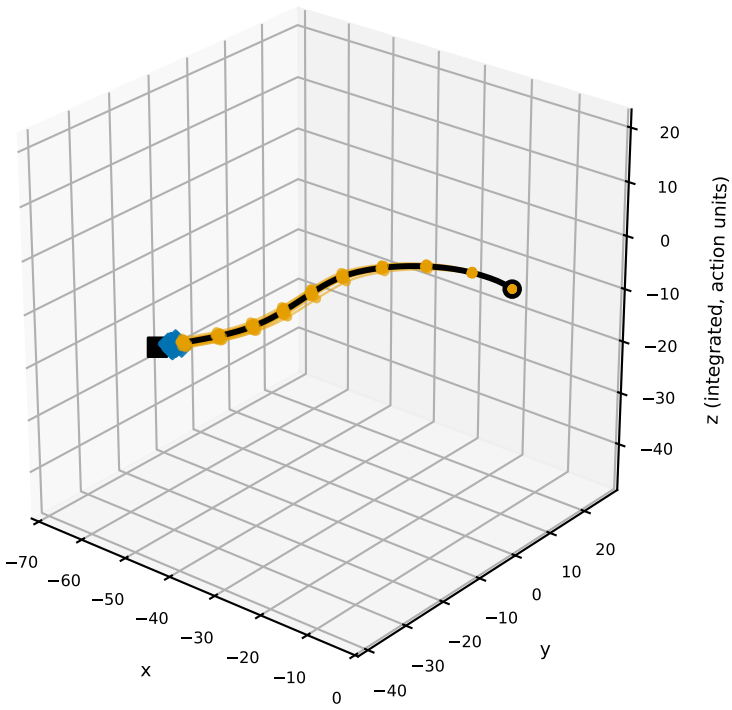
$\pi_{0.5}$ flow, typical state (sep 3.6)
path length 90.8; tube width mean 0.57 / max 1.98



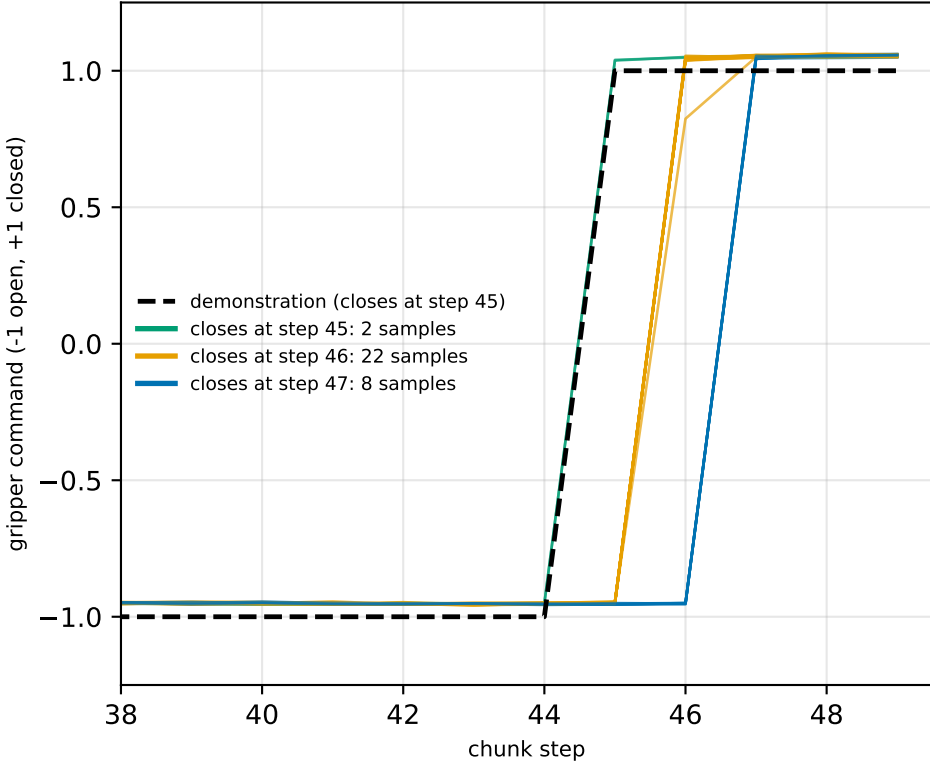
$\pi_{0.5}$ flow, widest state (sep 0.0)
path length 69.2; tube width mean 0.87 / max 4.78



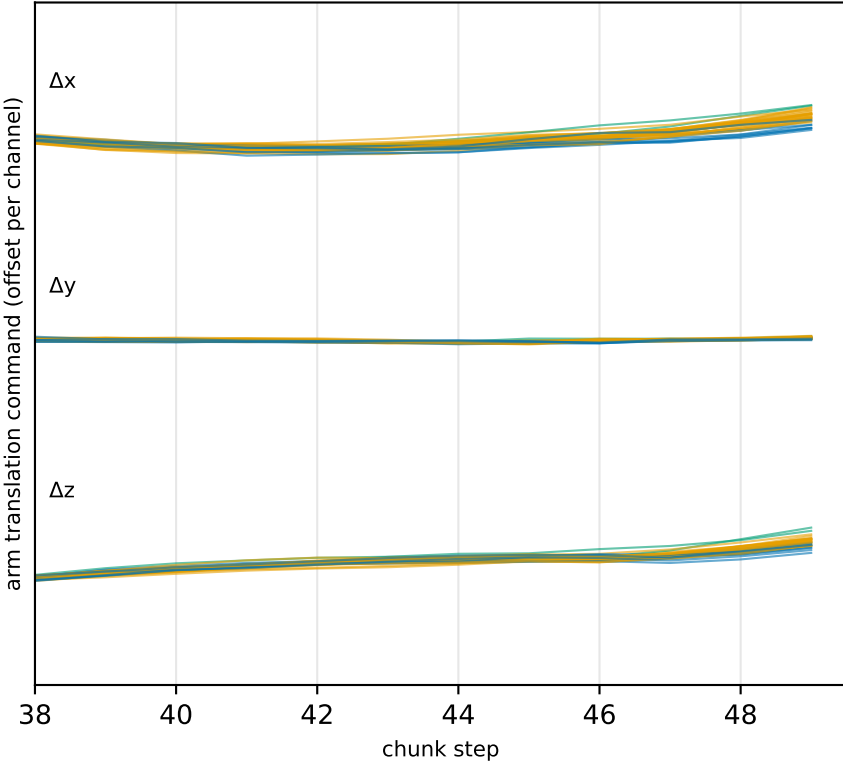
$\pi_{0.5}$ flow, most separated state in 3000 (sep 257, K=32)
= episode 176, frame 176: descending onto the mug
orange = gripper open, blue = closed (♦ = close step): step 46 in 22 samples, step 47 in 10 samples
path length 76.9; tube width mean 0.43 / max 1.30



(4) gripper channel of the 32 sampled chunks at the most separated state
all open then closed; the only difference is the flip step (45 / 46 / 47)



(5) arm channels of the same 32 chunks, same colors (each channel centred)
early-flip vs late-flip groups differ by 0.020 RMS on the arm; within-group scatter 0.015



(6) the same gripper channel with the observation shifted by -1 / +1 / +2 frames
the flip moves one step per frame; the ± 1 -step jitter stays

